

DOHA INTL, Qatar

WMO: 411700

Lat: 25.261N

Lon: 51.565E

Elev: 11

StdP: 101.20

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	13.8	-1.0	3.5	28.3	1.2	4.1	28.2	9.4	19.7	8.7	19.9	3.4	300	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	44.3	22.3	43.1	22.5	42.0	23.0	31.2	35.3	30.7	35.1	30.2	34.9	6.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
30.2	27.5	34.1	29.8	26.9	34.1	29.1	25.7	34.1	106.4	35.7	104.2	34.8	101.5	35.1	33.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	9.0	8.0	10.5	47.2	1.5	0.9	9.4	47.9	8.5	48.5	7.7	49.0	6.6	49.7		
(5)			6.6	32.1	1.7	0.6	5.4	32.5	4.4	32.9	3.4	33.2	2.1	33.6	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.8	18.7	19.8	23.1	28.0	33.7	36.2	37.1	36.4	34.4	31.0	25.7	20.8
	DBStd	7.01	2.33	2.47	2.62	2.75	2.45	2.14	1.81	1.69	1.78	1.86	2.50	2.45
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	41	23	11	1	0	0	0	0	0	0	0	0	7
	CDD10.0	6858	270	276	406	539	734	787	840	819	731	651	471	336
	CDD18.3	3857	34	53	148	289	476	537	582	561	481	392	221	84
	CDH23.3	54799	64	155	866	3144	7259	8924	9845	9322	7661	5417	1873	269
	CDH26.7	36166	6	28	247	1438	4823	6526	7365	6842	5263	3017	583	28
Wind	WSAvg	3.8	3.6	4.1	4.3	4.1	4.2	4.4	4.0	3.4	3.3	3.1	3.4	3.4
Precipitation	PrecAvg	72	12	18	21	5	0	0	0	0	0	1	5	13
	PrecMax	261	102	130	142	40	9	0	0	1	0	17	110	155
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	60	20	26	34	8	1	0	0	0	0	3	18	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	30.0	34.7	39.2	44.1	46.1	46.1	45.2	42.9	39.9	34.2	29.1
		MCWB	17.3	17.5	18.7	20.0	21.0	21.8	22.3	23.3	22.9	22.0	20.3	18.8
	2%	DB	25.0	26.8	31.2	36.2	42.0	44.1	44.2	43.2	41.0	37.9	32.2	27.2
		MCWB	17.4	17.1	18.1	19.6	21.4	22.0	22.2	24.1	23.6	22.7	21.6	19.7
	5%	DB	23.8	25.0	29.1	34.8	40.2	42.9	43.1	41.9	39.8	36.1	31.0	26.0
		MCWB	17.4	17.1	17.4	19.5	21.2	21.8	22.6	25.0	24.4	23.7	22.0	19.3
	10%	DB	22.4	23.8	27.3	33.0	38.9	41.2	41.9	40.2	38.2	34.9	29.9	24.9
		MCWB	17.1	16.9	17.4	19.6	21.1	22.0	23.3	26.3	25.7	24.3	21.8	19.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	20.8	22.2	24.4	27.7	29.8	31.7	32.1	30.9	29.0	25.9	23.1
		MCDB	22.9	23.7	26.0	29.7	31.9	33.7	35.6	35.8	35.5	33.4	29.1	25.3
	2%	WB	20.1	20.1	21.4	23.5	26.7	28.8	30.9	31.4	30.2	27.9	24.8	21.9
		MCDB	22.4	23.0	25.2	29.2	32.2	33.7	35.0	35.4	34.9	32.2	28.9	24.8
	5%	WB	19.2	19.4	20.7	22.8	25.7	28.1	30.4	31.0	29.7	27.2	24.0	21.0
		MCDB	21.9	22.5	24.7	28.8	32.5	33.4	34.7	35.2	34.5	31.8	28.4	24.1
	10%	WB	18.3	18.7	20.1	22.2	24.8	27.2	29.7	30.6	29.2	26.6	23.2	20.2
		MCDB	21.4	21.9	24.3	28.5	32.5	33.6	34.6	35.2	34.2	31.5	27.7	23.5
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.5	8.4	9.4	10.4	10.6	9.5	8.6	8.6	8.3	7.1	7.0
		MCDBR	8.1	9.0	10.8	11.7	12.0	12.1	11.4	10.8	10.3	10.0	8.2	7.9
		MCWBR	3.8	4.0	4.6	5.0	5.7	6.2	6.6	6.3	5.9	5.5	4.1	3.7
	5% WB	MCDBR	6.7	7.0	8.2	9.8	11.0	10.5	8.8	8.1	8.0	8.5	7.5	6.9
		MCWBR	3.9	3.7	4.2	4.7	6.1	6.7	6.0	5.0	4.8	4.9	4.2	3.7
Clear-Sky Solar Irradiance	taub	0.503	0.551	0.617	0.664	0.654	0.630	0.686	0.647	0.569	0.511	0.505	0.483	
	taud	1.892	1.782	1.644	1.593	1.595	1.636	1.592	1.673	1.800	1.916	1.936	1.955	
	Ebn at Noon	730	728	706	687	692	705	666	690	734	753	723	729	
	Edh at Noon	175	209	252	271	270	258	269	247	213	180	166	158	
All-Sky Solar Radiation	RadAvg	3.73	4.51	5.20	5.80	6.61	6.96	6.59	6.32	5.85	5.00	3.93	3.49	
	RadStd	0.22	0.24	0.32	0.41	0.18	0.18	0.17	0.15	0.11	0.12	0.19	0.23	

Historical Trends

DBAvg	Heating		Cooling			Degree-Days				
	99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	+0.65	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	+254	+239
Regional (3 neighbors)	+0.60	N/A	-1.48	+0.75	N/A	N/A	N/A	N/A	+238	+224

Nomenclature: See separate page